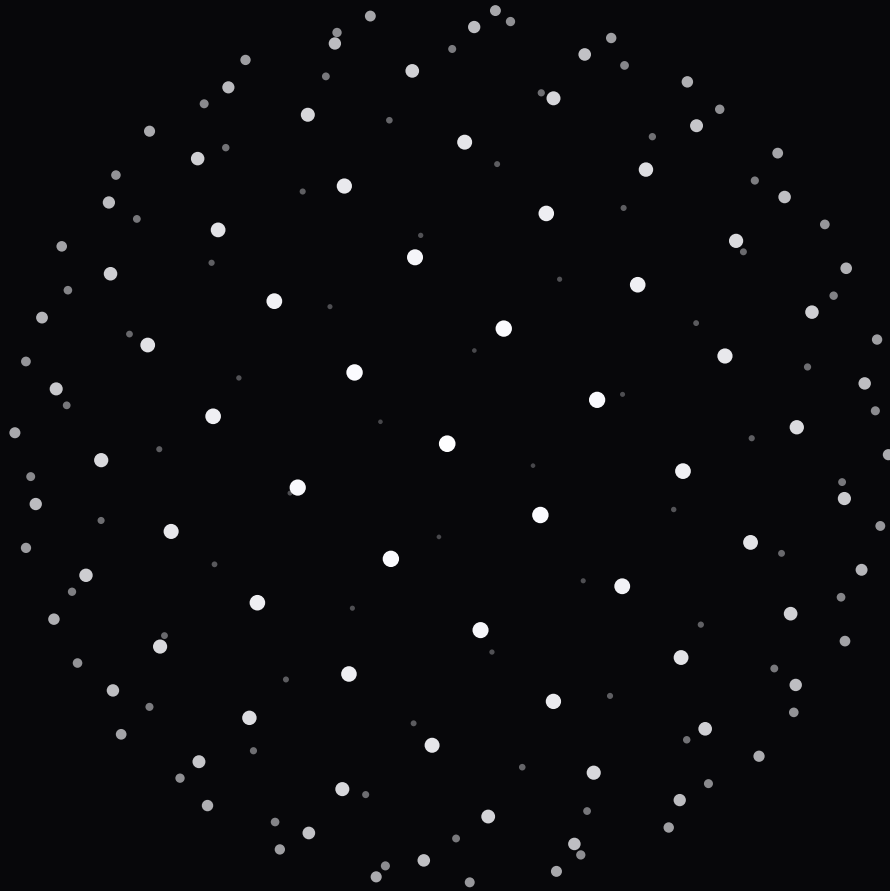

PROJECT REPORT

Ghost Customer AI

The AI that becomes your customer before your customer becomes your problem.



Executive Summary

Ghost Customer AI is a multi-agent customer-simulation platform. Instead of waiting for real users to reveal what is broken on a website, it generates hundreds of AI-powered virtual customers - 'ghosts' - that browse the site, evaluate pricing and onboarding, ask buying questions, and report where they would get confused, hesitate, or abandon. The platform then quantifies conversion risk, churn risk, support gaps and revenue leaks, and produces a board-ready executive report - all before a single real customer arrives.

The product is deliberately not a chatbot. A chatbot answers a customer after a problem occurs; Ghost Customer AI becomes the customer and surfaces the problem proactively, across an entire simulated market at once. It is built to run end-to-end with zero API keys on a deterministic engine, and to upgrade transparently to live AI when keys are present.

1. Problem Statement

Businesses typically discover experience problems only after they have already cost money: negative reviews, support tickets, cancellations, refund requests, and silent drop-off. Conversion blockers, unanswered sales objections, thin documentation and unclear pricing are invisible until traffic exposes them - by which point acquisition spend is wasted.

Existing AI tooling is reactive. Chatbots, support assistants and FAQ bots respond once a user complains. None of them simulate the customer's experience ahead of launch, and none predict which segments will churn or why a deal stalls.

2. Solution Overview

Ghost Customer AI inverts the model. It crawls and understands a target website, generates a diverse population of realistic customer personas, and runs each persona through the full funnel (Landing, Pricing, Features, Onboarding, Support). Specialized agents then act as a prospective buyer and as a difficult support customer to stress-test the site. The results are aggregated into quantified risks and a prioritized set of recommendations.

The user experience centers on a live 'war room' where the swarm is visualized in real time as a rotating field of customer nodes, followed by an interactive insights dashboard and an exportable executive report.

3. Product Features

- Customer Swarm Simulation - 12 to 500 AI personas across 10 archetypes, each simulating a complete buying journey and returning interest, confusion, purchase probability, support dependency, churn risk and a verdict (Convert / Maybe / Churn Risk / Bounce).
- Website Intelligence - extracts value propositions, calls-to-action, pricing tiers, FAQs, trust signals and a 0-100 content clarity score.
- Sales Objection Mining - surfaces the exact buyer questions the site never answers, scored by impact and affected roles.
- Support Stress Test - measures FAQ and documentation coverage against hard scenarios

and flags where support dependency and churn rise.

- Revenue Leak Detection - names each leak, its cause, the percentage of customers affected, the estimated conversion loss and the recommended fix.
- Churn Prediction - identifies which segments will leave and why (Pricing, Onboarding, Missing Features, Poor Support, Complexity).
- Competitor Battle Arena - scores your site against a rival across dimensions and shows which customer segments prefer which product.
- Pricing Time Machine - a constant-elasticity demand model that finds the revenue-maximizing price and predicts how each segment reacts to a change.
- Executive Report - a narrative summary with prioritized, effort-vs-impact recommendations, exportable to PDF and Markdown.

4. Multi-Agent Architecture

The platform is built around a canonical eight-agent graph. Website Analyzer feeds Persona Generator, which feeds the Customer Simulation Swarm. The Sales Agent and Support Agent run as parallel branches after simulation and fan back into the Revenue Leak Agent, which is followed by the Insight Agent and finally the Report Generator.

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Website Analyzer -> Persona Generator -> Customer Simulation Swarm
                    -> Sales Agent   ||   Support Agent
                    -> Revenue Leak Agent -> Insight Agent -> Report Generator
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Language-heavy stages (site understanding and report narrative) are handled by the LLM, where reasoning adds the most value. Per-persona scoring stays in a fast, grounded, deterministic engine, so a 500-ghost swarm never fans out into 500 fragile model calls and every number traces back to the same detected gaps.

5. Technical Architecture & Stack

The product ships as a full-stack TypeScript application (Next.js 15, React 19, Tailwind, Framer Motion, Recharts). Its API routes implement the multi-agent orchestrator and stream results to the browser as newline-delimited JSON, driving the live war-room UI. A parallel Python engine (FastAPI + LangGraph) implements the identical agent graph and contract, and can be enabled via an environment variable.

AI is provided by Gemini 2.5 Flash; crawling by Firecrawl with a plain-fetch fallback; optional persistence by Supabase Postgres with pgvector. Crucially, all external services are optional: with no keys the app runs on a deterministic, seeded mock engine that produces rich, reproducible results, guaranteeing the demo always works.

Deployment: the Next.js app (frontend plus its API-route backend) deploys to Vercel as one unit; the optional Python engine deploys to Render or Railway. Docker and docker-compose files are provided for fully offline, local operation.

6. Data Model & API

Core domain types include WebsiteAnalysis, Persona, SimulationResult, Insights, CompetitorAnalysis, PricingSimulation and ExecutiveReport, shared verbatim between the TypeScript and Python engines. The Supabase schema defines users/profiles, projects, crawled_pages and documents (with vector embeddings), personas, simulations, insights, reports and competitor_analysis, with row-level security and pgvector indexes.

- POST /api/run - streams the full multi-agent run as NDJSON RunEvents.
- POST /api/project, /api/crawl, /api/generate-personas - project + analysis utilities.
- POST /api/competitor-analysis, /api/pricing-simulation - standalone analyses.
- GET /api/insights, /api/report, /api/health - results and status.

7. Differentiation

Most AI products in this space are reactive: chatbots, support assistants and FAQ bots that engage after a customer raises an issue. Ghost Customer AI is proactive and population-scale: it simulates the whole market before launch. The closest analogies are chaos engineering and digital-twin technology applied to customer behavior rather than infrastructure.

8. Roadmap

- Deeper LLM reasoning mode that routes a sample of personas through full model reasoning.
- Mission-control dashboard layout and an interactive revenue-leak 'fracture map' of the site.
- Real document/PDF ingestion and competitor multi-page crawling.
- Persisted run history and team collaboration via Supabase Auth.

9. Conclusion

Ghost Customer AI turns customer discovery from a lagging indicator into a leading one. By becoming hundreds of customers on demand, it lets a business see - and fix - conversion blockers, churn risks and revenue leaks before real customers ever experience them. It is production-quality, demo-proof, and architecturally honest about where AI adds value versus where deterministic logic keeps results fast and reproducible.

Source code: github.com/garvbahl37-gif/GhostAI